ALGEBRA ON ONE PAGE

Polynomial long division and the ring $\mathbb{Q}[x]$

Division with remainder gives Euclid, Bezout and unique factorisation for polynomials, as for whole numbers.

1 / THE RULE AND THE PICTURE

Long division on coefficients removes the highest power again and again

	x^4	x^3	x^2	x	1	
			2	-2	0	quotient $2x^2-2x$
x^2+x+1	2	0	0	0	1	
$-$	2	2	2			$2x^2\cdot b$
		-2	-2	0	1	
$-$		-2	-2	-2		$-2x\cdot b$
				2	1	remainder $2x+1$

Dividing a by b asks for a quotient q and a remainder r with

$$a = q\,b + r, \qquad \deg r < \deg b$$

Each step cancels the leading term of what is left by subtracting a multiple of b. Here factorising out $2x^4+1$ subtracting $2x^2\cdot b$ cancels the x^4 term and leaves $-2x^3-2x^2+1$, and then subtracting $(-2x)\cdot b$ cancels the x^3 term and leaves $2x+1$, which has lower degree than b, so the process stops. The multipliers $2x^2$ and $-2x$ add up to the quotient $q=2x^2-2x$, and $r=2x+1$ is the remainder. The columns of the picture are the coefficients of $x^4,\dots,1$. Over $\mathbb{Q}$ the method always works, because cancelling $c\,x^k$ needs the multiplier $\frac{c}{\beta}\,x^{k-\deg b}$, where β is the leading coefficient of b, and $\frac{c}{\beta}$ is again rational.

2 / EUCLID'S ALGORITHM

Dividing the divisor by the remainder, until the remainder is zero, finds the gcd

Repeating the division on b and the remainder gives

$$\begin{aligned} 2x^4+1 &= (2x^2-2x)(x^2+x+1)+(2x+1)\\ x^2+x+1 &= (\tfrac{1}{2}x+\tfrac{1}{4})(2x+1)+\tfrac{3}{4}\\ 2x+1 &= (\tfrac{8}{3}x+\tfrac{4}{3})\cdot\tfrac{3}{4}+0 \end{aligned}$$

and the last nonzero remainder $\frac{3}{4}$ is a constant, which divides everything, so the gcd is 1 up to a constant factor and a, b are coprime. The same steps work for whole numbers, and the table sets the two side by side:

	whole numbers $\mathbb{Z}$	polynomials $\mathbb{Q}[x]$
division with remainder	$a=qb+r$ with $\lvert r\rvert<\lvert b\rvert$	$a=qb+r$ with $\deg r<\deg b$
can be divided by	±1	nonzero constants
primes	$2,3,5,\dots$	irreducible polynomials
Euclid, an example	$126=4\cdot30+6$, $30=5\cdot6$: gcd 6	$x^3-1=x(x^2-1)+(x-1)$, $x^2-1=(x+1)(x-1)$: gcd $x-1$

3 / READING EUCLID BACKWARDS

The gcd is a combination of the two polynomials

Substituting $\frac{3}{4}=b-(\frac{1}{2}x+\frac{1}{4})\,r$ and $r=a-(2x^2-2x)\,b$ gives

$$1 = m\,a+n\,b, \quad m=-\tfrac{2x+1}{3}, \quad n=\tfrac{4x^3-2x^2-2x+4}{3}$$

multiplying out giving prime divisor property: if an irreducible p divides ab but not a, then $1=mp+na$, so $b=mpb+n(ab)$ is divisible by p.

4 / THE WIDER VIEW: $\mathbb{Q}[x]$ IS A RING

Everything above comes from division with remainder

Polynomials with rational coefficients can be added, subtracted and multiplied without leaving the set, with 0 and 1, so $\mathbb{Q}[x]$ is a ring. It is not a field, since $1/x$ is not a polynomial and only the nonzero constants have inverses. Because the degree measures size and division with remainder exists, $\mathbb{Q}[x]$ is a Euclidean ring like $\mathbb{Z}$, and gcds, Bezout and unique factorisation follow. The same holds over any field, such as $\mathbb{F}_p$, $\mathbb{R}$ or $\mathbb{C}$, since we must divide by leading coefficients; $\mathbb{Z}[x]$ fails, as dividing x by $2x$ needs the coefficient $\frac{1}{2}$. Enlarging the coefficients simplifies the primes: over $\mathbb{Q}$ there are irreducibles of every degree (x^n-2), over $\mathbb{R}$ only of degree 1 or 2 (x^2+1), and over $\mathbb{C}$ only of degree 1.

5 / FROM RING TO FIELD

Congruence modulo an irreducible polynomial gives a field, and Euclid finds the inverses

Modulo x^2-2 every polynomial leaves a remainder $a+bx$ with $x^2\equiv 2$, which is $\mathbb{Q}(\sqrt{2})$ with basis $1,\sqrt{2}$ and dimension 2; for example $x^4+3x+1\equiv 3x+5$, the number $5+3\sqrt{2}$. The inverse of $1+x$ comes from $x^2-2=(x-1)(x+1)-1$, which says $(x+1)(x-1)\equiv 1$, so $\frac{1}{1+\sqrt{2}}=\sqrt{2}-1$: the twin trick of the adding-a-number sheet is this step of Euclid's algorithm. The modulus must be irreducible, and x^2-2 is, since a factor $x\pm a$ would need $a^2=2$ with a rational. In the same way $\mathbb{Q}[x]/(x^2-x+1)$ is $\mathbb{Q}(\Phi)=\mathbb{Q}(\sqrt{-3})$, $\mathbb{R}[x]/(x^2+1)$ is $\mathbb{C}$, and $\mathbb{F}_2[x]/(x^2+x+1)$ is the field with four elements.

ASIDE / TRY THESE

Show $\gcd(x^2-1,\,x^2-2x+1)=x-1$. The remainder of x^3+2x+1 on division by $x-1$ is 4, the value at $x=1$.

LINKS TO OTHER SHEETS

Adding a new number and field extensions: $\mathbb{Q}(\sqrt{2})$ and cyclotomic fields; the fractions sheets use linear polynomials $bx-a$.

Convention. $\mathbb{Q}[x]$ polynomials with rational coefficients; $\deg$ degree; all divisions and the identity $1=ma+nb$ were checked symbolically.

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